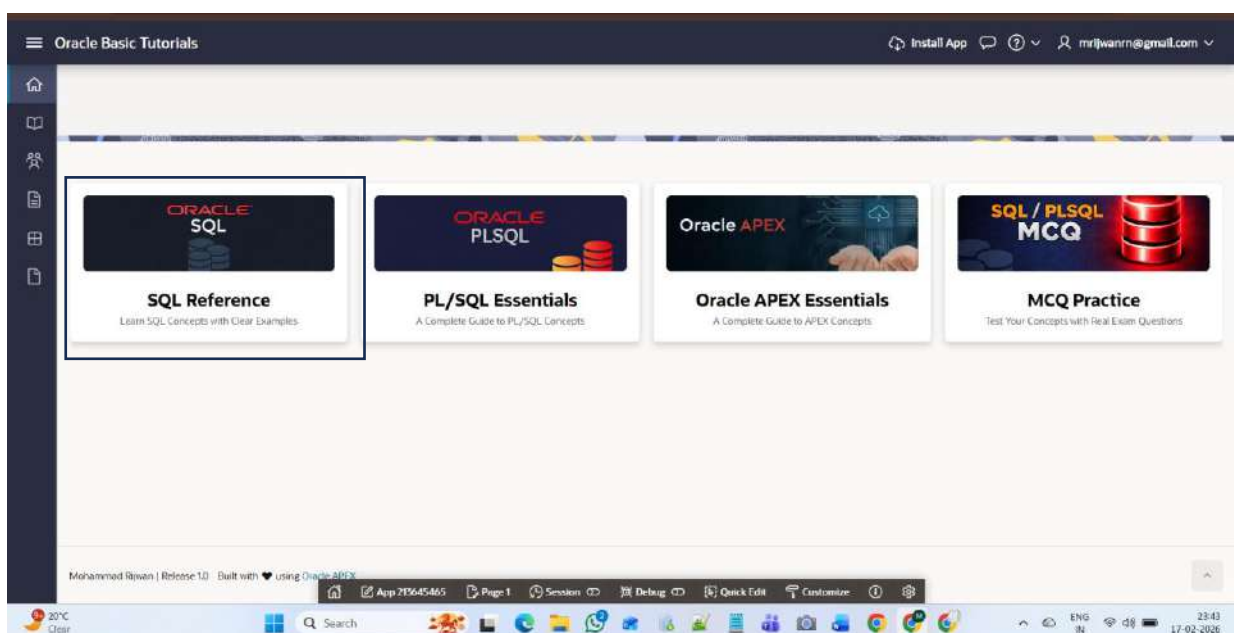


1. HOME PAGE

Overview: The Oracle Basic Tutorials application allows users to open tutorials using two different navigation methods:

1. By selecting a tutorial card from the Home page
2. By using the Hamburger Menu and navigating through the Course section

This dual navigation system improves usability and provides flexibility for different types of users.



Method 1: Open Tutorial by Selecting a Card

Location: Home Page

❖ Available Cards:

- SQL Reference
- PL/SQL Essentials
- Oracle APEX Essentials
- MCQ Practice

➤ **Steps:**

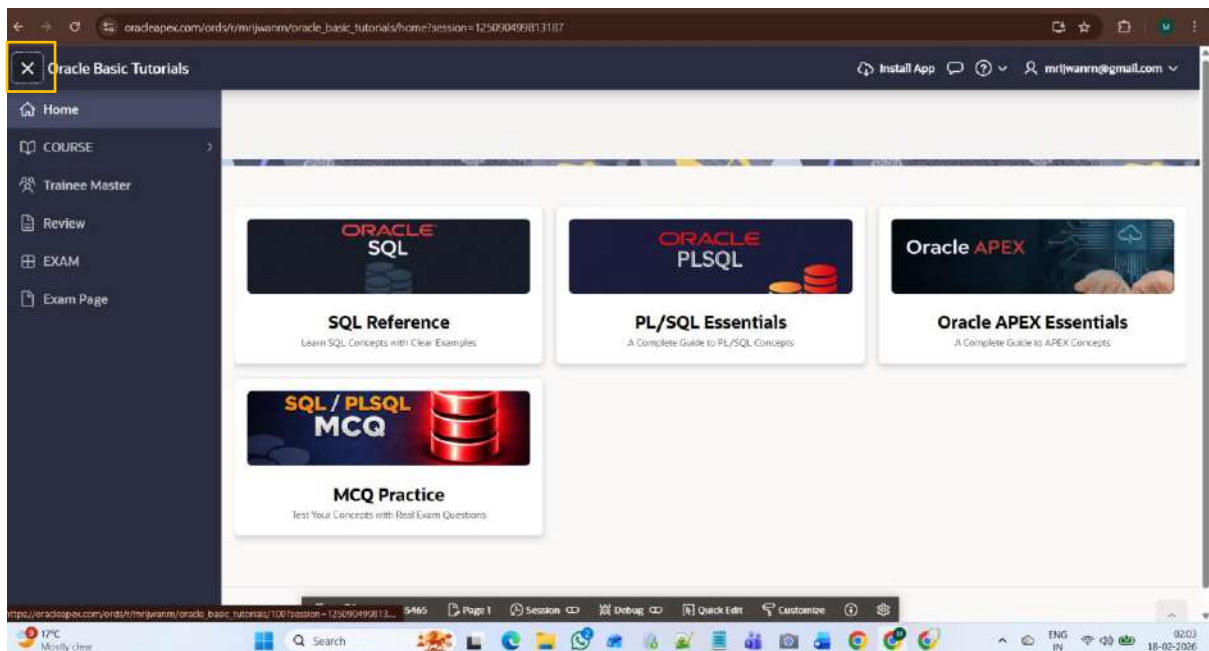
1. Open the application.
2. On the Home page, view the available tutorial cards.
3. Click on any card.

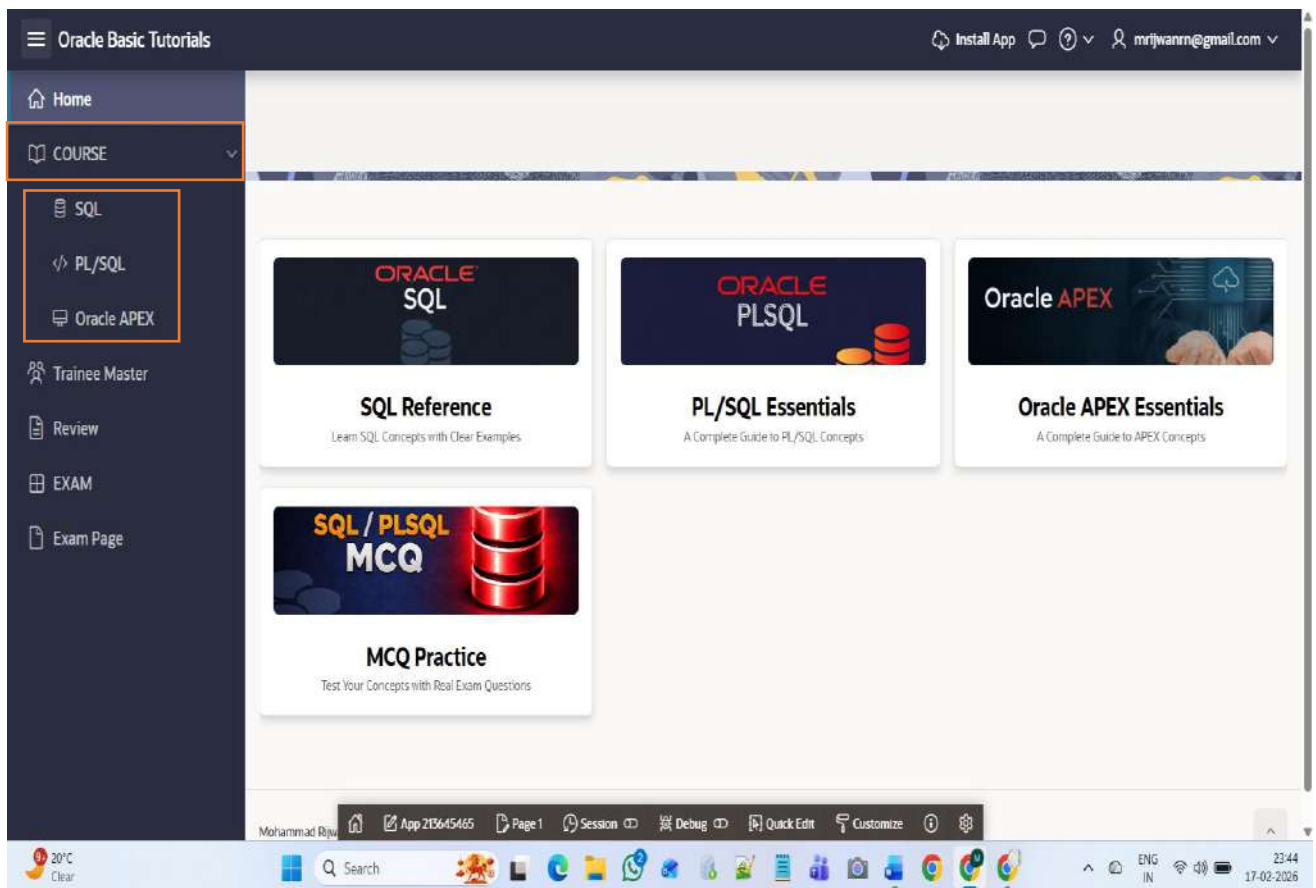
➤ **System Behaviours**

- When a card is clicked, the system redirects the user to the corresponding tutorial page.
- The navigation is handled dynamically using page URLs.
- Each card is linked to its respective tutorial module.

➤ **Advantages:**

- Quick access
- User-friendly visual interface
- Ideal for beginners





Method 2: Open Tutorial Using Hamburger Menu

Location: Top-left Hamburger Icon (≡)

➤ Steps:

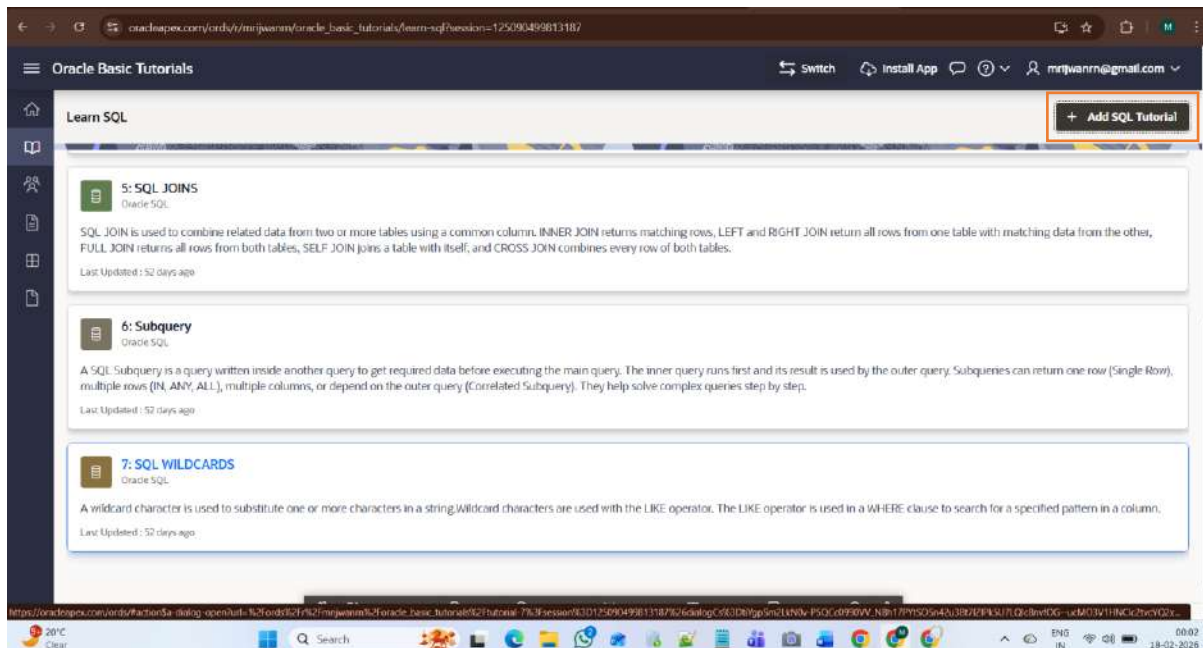
1. Click the Hamburger icon in the top-left corner.
2. The sidebar menu will open.
3. Click on the **COURSE** section.
4. Select the required category:
 - SQL
 - PL/SQL
 - Oracle APEX

➤ System Behaviours:

- The system redirects the user to the selected course listing page.
- This navigation method uses structured menu-based routing.

➤ Advantages:

- Organized navigation
- Direct category access
- Suitable for regular or advanced users



After selecting a course (either from a Home page card or from the Hamburger → Course menu), the system opens a Course Dashboard Page.

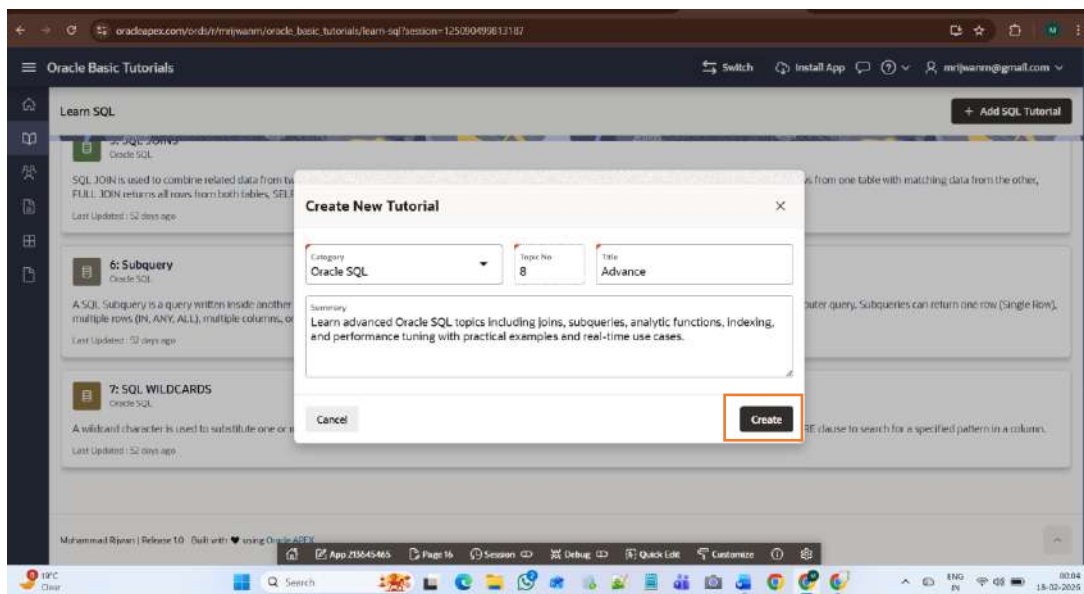
3. Course Dashboard Page

➤ When User Clicks:

- SQL Card → Opens SQL Dashboard
- PL/SQL Card → Opens PL/SQL Dashboard
- Oracle APEX Card → Opens APEX Dashboard

➤ What Appears on Dashboard:

- The page displays:
 - Topic Number
 - Topic Title
 - Summary
 - Last Updated Date
- Each tutorial is displayed in Card Format.

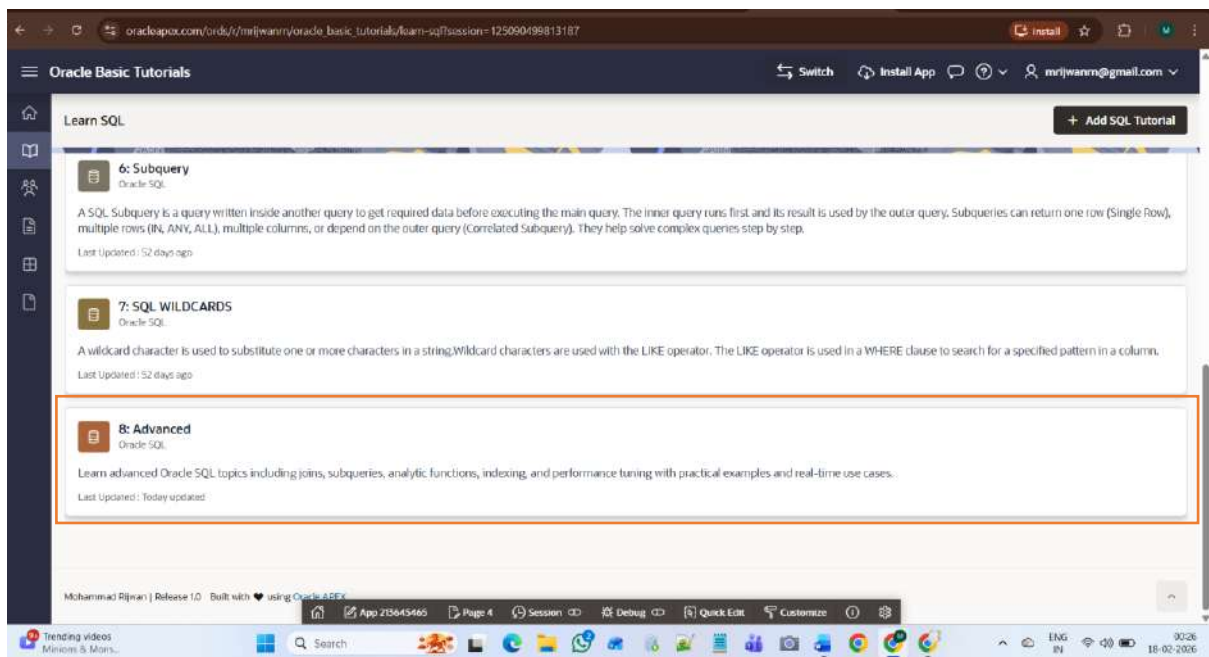


4. Adding a New Tutorial

On each Course Dashboard page, a button is available: + Add SQL Tutorial
(or PL/SQL / Oracle APEX depending on category)

➤ Steps to Add New Tutorial

1. Click **Add Tutorial** button
2. A Modal Dialog Form opens
3. The form requires:
 - Category (Auto selected)
 - Topic Number (Auto generated)
 - Title (Required)
 - Summary (Required)



5. Creating the Tutorial

After filling required fields:

1. Click Create button
2. System performs:
 - Insert record into database
 - Auto assigns next topic number
 - Saves category-wise
3. Modal dialog closes
4. Dashboard refreshes automatically
5. Newly created tutorial appears as a new card

Oracle Basic Tutorials

Learn SQL

- 1: SQL Overview**
Oracle SQL
SQL (Structured Query Language) is the standard language used to store, retrieve, and manipulate data in a database.
Last Updated: 52 days ago
- 2: Sub Languages in SQL**
Oracle SQL
This topic covers the five main sublanguages of SQL: DDL, DML, TCL, DCL, and Managing permissions.
Last Updated: 52 days ago
- 3: Constraints**
Oracle SQL
SQL Constraints are rules applied on table columns to ensure data accuracy.
Last Updated: 52 days ago
- 4: FUNCTION IN SQL**

Tutorial - 1

Introduction to SQL

SQL stands for **Structured Query Language**. It is the standard language used for storing, manipulating, and retrieving data in a database. SQL became a standard of the **American National Standards Institute (ANSI)** in 1986 and the **International Organization for Standardization (ISO)** in 1987. SQL is a standard language for accessing and manipulating databases.

What is SQL?

- SQL stands for Structured Query Language
- SQL lets you access and manipulate databases
- SQL became a standard of ANSI in 1986
- SQL became a standard of ISO in 1987

What Can SQL Do?

- Execute queries against a database
- Retrieve data from a database

Oracle Basic Tutorials

Learn PL/SQL

- 1: Introduction to PL/SQL**
PL/SQL
PL/SQL stands for **Procedural Language / Structured Query Language**. PL/SQL is a block-structured programming language that combines SQL statements with procedural features like conditions and loops. It is mainly used to write business logic directly inside the database.
Last Updated: 51 days ago
- 2: Programming Fundamentals**
PL/SQL
This module explains the basic programming concepts used in PL/SQL, such as variables, loops, and conditions.
Last Updated: 51 days ago
- 3: Control Structures**
PL/SQL
This module explains how to control the flow of execution in a PL/SQL program using control structures like IF, LOOP, and GOTO.
Last Updated: 51 days ago
- 4: Database Interaction**
PL/SQL

PLSQL Module 1

PL/SQL Basics

PL/SQL stands for **Procedural Language / Structured Query Language**. It is Oracle's extension of SQL that allows us to write **procedural code** inside the database.

1. Introduction to PL/SQL

PL/SQL is a block-structured programming language that combines **SQL statements** with **procedural features** like conditions and loops. It is mainly used to write business logic directly inside the database.

2. Advantages of PL/SQL

- Improves performance by reducing network traffic
- Supports variables, loops, and conditions
- Better security for database logic
- Easy integration with SQL
- Reusable code using procedures and functions

3. PL/SQL Architecture

PL/SQL code is processed by the **PL/SQL Engine**.

Oracle Basic Tutorials

Learn APEX

- 1: Oracle APEX Introduction**
APEX
This module introduces Oracle Application Express (APEX), a low-code development platform used to build **secure, scalable, and data-driven web applications**.
Last Updated: 51 days ago
- 2: APEX Workspace & Environment**
APEX
This module explains how to access Oracle APEX, create a workspace, and set up the environment.
Last Updated: 51 days ago
- 3: APEX Application Basics**
APEX
This module explains how to create and manage Oracle APEX applications.
Last Updated: 51 days ago
- 4: Page Designer Deep Dive**
APEX

APEX Module 1

Oracle Apex Introduction

This module introduces **Oracle Application Express (APEX)**, a low-code development platform used to build **secure, scalable, and data-driven web applications**.

1. What is Oracle APEX?

Oracle APEX is a low-code development tool provided by Oracle that allows developers to create web applications using **SQL and PL/SQL**. APEX applications run inside the Oracle Database and do not require external application servers.

2. History & Versions of APEX

Oracle APEX was originally known as **HTML DB**. It was later renamed to Oracle Application Express and is continuously improved with every release.

3. APEX Architecture

Oracle APEX works on a **three-tier architecture**:

- Web Browser (Client)
- Web Listener (ORDS)

5. Course Tutorial – Detailed View (Drawer Page)

When the user clicks on any course tutorial from the dashboard (SQL, PL/SQL, or Oracle APEX), the system opens the selected tutorial in a drawer panel on the right side of the screen.

The drawer page displays the complete detailed content of the selected tutorial, including:

- Tutorial Title
- Module Name
- Section Headings
- Detailed Explanation
- Structured Content (bullet points, paragraphs, highlights)

The system dynamically fetches the tutorial data from the database using the selected Topic ID. The main dashboard remains visible in the background while the detailed content appears in the sliding drawer.

The drawer includes a close (X) button that allows the user to close the detailed view and return to the course dashboard without reloading the page.

This functionality works consistently for all courses:

- SQL Tutorials
- PL/SQL Tutorials
- Oracle APEX Tutorials

Each tutorial opens its own detailed content based on the selected topic.

Review

Course Reports

Search: All Test Columns

Go: Primary Report

Actions

Topic	Category	Duration	Test Rating	Feedback
SQL Overview	Oracle SQL	1 Day	5.5	Good
Sub-Languages in SQL	Oracle SQL	7 Days	4	Very Good
Constraints	Oracle SQL	1 Day	3.5	Good
FUNCTION IN SQL	Oracle SQL	6 Days	3.5	Good
SQL JOINS	Oracle SQL	11 Days	4	Very Good
Subquery	Oracle SQL	17 Days	3	Need To Improve
SQL WILDCARDS	Oracle SQL	1 Day	3.5	Good
Introduction to PL/SQL	PL/SQL	5 Days		
Programming Fundamentals	PL/SQL	10 Days		
Control Structures	PL/SQL	7 Days		
Database Interaction	PL/SQL	2 Days		
Cursor	PL/SQL	3 Days		
Records & Collections	PL/SQL	1 Day		
Exception Handling	PL/SQL	1 Day		
Procedures & Functions	PL/SQL	1 Day		
Triggers & Dynamic SQL	PL/SQL	0		
Packages	PL/SQL	1 Day		
Oracle APEX Introduction	APEX	0		

1 rows selected

App: Oracle Basic Tutorial Page 12 | Version: 1.0 | Testing: 1.0 | Quickstart | Customize

Total: 30

6. Review Module – Test-Based Course Evaluation

Overview

The **Review Page** provides a structured report of course performance based on user test results and feedback.

This module helps analyse how each tutorial or topic is performing in terms of:

- Learning duration
- Test rating
- User feedback
- Category performance

6.1 Purpose of the Review Page

The Review module is designed to:

- Monitor tutorial effectiveness
- Track user performance
- Evaluate topic quality
- Identify improvement areas
- Provide feedback analysis

It acts as an analytical dashboard for course performance.

6.2 Course Reports Section

The page displays a **Primary Report (Interactive Report)** containing the following columns:

1. Title

Displays the name of the tutorial or topic.

Example:

- SQL Overview
- Sub Languages in SQL
- Subquery
- Introduction to PL/SQL
-

2. Category

Indicates which course the tutorial belongs to:

- Oracle SQL
- PL/SQL
- Oracle APEX

3. Duration

Represents the number of days spent or allocated for the topic.

Example:

- 1 Day
- 7 Days
- 17 Days

This helps evaluate the learning timeline.

4. Test Rating

Displays the performance rating of the topic based on test results.

Example:

- 3.5
- 4
- 3

Higher rating indicates better understanding and effectiveness.

5. Feedback

Shows qualitative feedback collected after the test.

Examples:

- Good
- Very Good
- Need To Improvement

This helps measure student satisfaction and topic clarity.

6.3 Functional Features

The Review page includes:

- Search functionality (Search All Text Columns)
- Report filtering options
- Sorting capability
- Actions menu for exporting or customizing reports
- Editable option (if enabled)

6.4 How Review Process Works

1. User completes course topic.
2. User attempts test or evaluation.
3. System calculates rating.
4. Feedback is recorded.
5. Data is stored in the database.
6. Review page fetches and displays updated report.

7. Header Controls & Application Information

7.1 Switch Button

The **Switch** button is located in the top-right section of the application header.

Functionality:

When the user clicks the **Switch** button:

- The system redirects the user back to the **Home Page (Dashboard)**.
- It allows quick navigation from any course or tutorial page.
- No data is modified during this action.
- It acts as a global navigation shortcut.

Purpose:

- Provides quick return access to the main dashboard.
- Improves user navigation flow.
- Ensures consistent access to the starting point of the application.

7.2 Install App Button

The **Install App** button is also located in the top-right header area.

Functionality:

When the user clicks the **Install App** button:

- The system initiates the Progressive Web App (PWA) installation process.
- The application can be installed on the user's device (desktop or mobile).
- Once installed, the application behaves like a native application.
- It can be accessed directly from the home screen or desktop shortcut.

Benefits:

- Offline-style experience
- Faster loading
- App-like interface
- Improved accessibility

7.3 Release Information (Footer Section)

At the bottom of the application, the footer displays release information.

Components Explained:

- Developer Name: Mohammad RIJWAN
- Release Version: 1.0
- Technology Used: Oracle APEX

Purpose:

- Identifies the application version.
- Provides developer attribution.
- Displays platform acknowledgment.
- Helps track application updates and future releases.

Final Application Summary

The Oracle Basic Tutorials application includes:

- Dual navigation system (Card & Sidebar)
- Dynamic course dashboards
- Modal-based tutorial creation
- Drawer-based detailed content view
- Header navigation controls (Switch, Install)
- Release tracking and version display

This ensures a complete, structured, and scalable tutorial management system built using Oracle APEX.